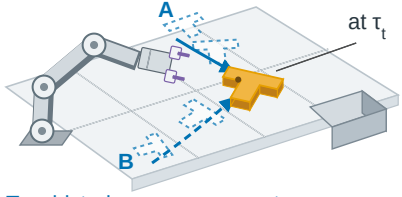
What must be remembered and predicted for interaction?

Conceptual illustration; all positions and trajectories are hypothetical

(a) Same pose, different motion



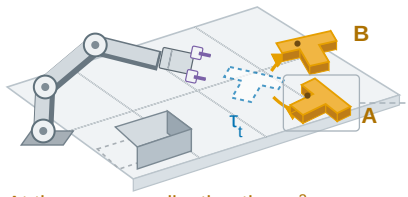
Two histories → one current pose

Past poses at −100, −50, 0 ms

A B

Current pose alone is ambiguous.

(b) Different interaction poses



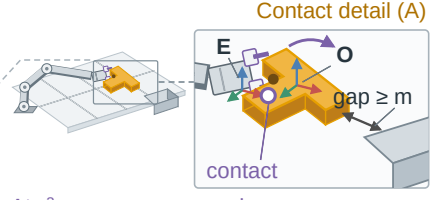
At the same application time τ^a

A / B → different position + rotation

Predict its SE(3) pose at that time.

Carry the motion history forward.

(c) Resolve contact geometry



At τ^a: pose error ≤ ε and gap ≥ m

Target: tolerance ε, exact SE(3).

Context: margin m ≫ ε, 3D flow.

O: target; E: end-effector frame